

Magnetotherapy

Definition:

- It is a therapeutic technique that consists in applying artificial magnetic fields on an affected part of the human body, controlling intensity and frequency.
- So we expose the affected body's area to a magnetic field similar to the Earth's one (intensity of 0,5 Gauss), but with artificial magnetic fields we can get **higher levels of frequency** and **application time**.

Biological Effects of Magnetic Fields:

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graph TD; A[Biological Effects of Magnetic Fields:] --> B[1. Cellular effects.]; A --> C[2. Effects on Organs and Systems.];
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1. Cellular effects.

2. Effects on Organs and Systems.

Cellular effects:

1. Enrich the oxygen in the cell.
2. Improve the utilization of oxygen.
3. Increasing the metabolism.
4. Enhance ATP production.
5. Increases energy on the cellular level.

Effects on Organs and Systems:

I. OSSEOUS TROPHIC EFFECT

The magnetic field, due to the piezoelectric property of the bone, makes better the architectural course of osseous trabeculas in damage zones, guiding them to the same direction. This means that better osseous callus is created.

In conclusion, a certain number of exposures to a low frequency magnetic field provoke the formation of an osseous callus of high-quality in a short time period.

Magnetic fields:

1. Collagen production increase.
2. Calcium deposit increase.
3. Modification of the osseous architectural process.
4. Stimulation of the hyaluronic acid synthesis.

Effects on Organs and Systems:

II. VASCULAR EFFECT

It has been observed that magnetic fields facilitate the capillary neoformation, the appearance of pericytes in capillaries, and the opening of precapillary sphincters. By:

1. There is a vasodilatation in a local level due to arterial circulation improvement.
2. There is an improvement of the oxygen partial pressure.

Effects on Organs and Systems:

III. ANALGESIC EFFECT

There is an evident analgesic effect, but the genesis is still unknown. Hypothesis:

- The production's increase of oligopeptides that constitutes the group of endorphins and enkephalins, being this effect at a central and peripheral level.

Effects on Organs and Systems:

IV. Autonomous Nervous System Effects:

- Increase of chemical mediator levels, as catecholamine (adrenaline, noradrenalin) or as acetylcholine.
- Depending on the orientation of the nodular ganglion and vagus nerve in a magnetic field, an increase or inhibition of the nervous impulse transmission is observed.
- This is due to the orientation produced by the magnetic field in the ions of the cellular membrane.

How a Magnetic Field is Produced:

1. An electrical current produces an associated magnetic field.
2. Inside natural magnets is crossed by microscopic electric currents which cause a magnetic field of certain intensity and direction due to the structure of these crystals.

Types of artificial magnets used:

1. **Not Stationary (Electro-magnets):** work only when connected to electricity.
 - Pulsed: (Pulsed generator connected to a coil → magnetic field → current flows in tissue).



Types of artificial magnets used:

- 2. **Stationary (Permanent magnets):** are fully charged with electric current and remain permanently magnetized.
(Earrings, bracelets, necklace or pellets)
- Unipolar: discrete magnets arranged with same pole facing patient's skin.
- Bipolar: both North & South poles facing patient's skin.



Application Norms:

1. The patient must NOT introduce metal objects in the solenoid because they can distort the magnetic field, not for expect an injurious effect, just for modification of magnetic field parameters.
2. The treatment usually response quickly, but it depends on the pathology, age, and other conditions of the patient. If after 10 sessions we do not see the patient is advanced along the curative process, it is better to stop the therapy because it means the patient is not receptive to the treatment.
3. The therapy time 30 minutes to 2 hours maximum.
4. It is better for the affected zone of the patient to be in direct contact with solenoid.
5. The sessions must be applied continuously. If it is necessary to accelerate the recovery's patient, we could apply two sessions at the same day but with a separation of 12 hours between them.
6. Electromagnetic fields therapy can be complemented with other therapies

Basic guidelines

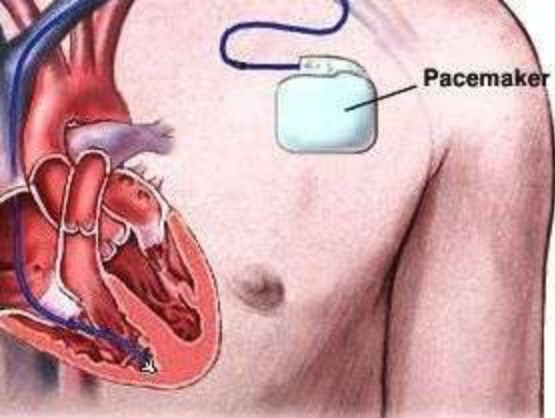
- All treatments should start at lower settings.
- Changes and reactions should be observed to allow adjustment of treatment settings.
- An increase of pain may occur, which is a positive indication that the patient is actually responding to the therapy [reaction phase]

Basic rules

- Increase of pain → lower setting
- Less pain stay with setting
- No change in condition go to higher setting
- Head areas & migraine shall never be treated higher than 1-3mT, 2Hz, 10 min
- Patient with allergies may also require low settings 1-3mT, 2-4Hz, 10-20 min
- Attention must be paid to the patient with severe circulatory disorders, pain can increase with low setting due to increased blood flow in peripheral vessels.

Main applications:

1. Spondylosis.
2. Muscle spasm.
3. Epicondylitis.
4. Sciatica
5. Trigeminal neuralgia.
6. Gout.
7. RA.
8. Recent trauma.
9. Fractures.
10. Trophic ulcers.
11. CRPS.
12. Dupuytren's contracture.
13. Osteoporosis.
14. Osteomyelitis acute & chronic.
15. Dysmenorrhoea & prostatitis.
16. Sinusitis.
17. Depression & Migraines.



dications:

1. **Existence of acute infection**, specially septic tuberculosis, because there is danger of dissemination.
2. **Hemorrhagic focus** (bleeding ulcers, intestinal etc), because of vasodilatation effect can provoke haemorrhage.
3. **Cardiac pacemaker**, because it can produce a malfunction.
4. **Pregnancy**, because it could provoke foetal damages.
5. **Tumours existence** because they can be disseminated.
6. Patients with **hydrocortisone treatment**.
7. In case of **vascular obstruction**, because the obstructive element can be liberated and obstruct another part of the body, provoking further damages.
8. Juvenile diabetes –during PMF treatment BGL should be monitored

